

Prediction of Carbon Monoxide (CO) Levels Using Machine Learning

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Abstract

Air pollution is a significant environmental and health concern globally. The ability to accurately monitor and predict air pollutant concentrations is vital for public health warnings. This report details a machine learning program aimed to analyze and predict air quality based on the sensor data from a heavily trafficked area in an Italian city. Using the UCI machine learning repository's 'Air Quality' dataset, we performed data cleaning, exploratory data analysis and feature engineering. We trained and evaluated the Generalized Linear Model (GLM) with a Gaussian family, which is a linear regression model. It is being used to predict CO(GT) value based on the features in the dataset. The model effectively predicts CO(GT) levels using various air quality parameters. The significant predictors suggest relationships between CO and other pollutants.

Introduction

Air quality is a big deal for our health and the planet. The WHO says ambient air pollution is linked to millions of premature deaths every year, mainly due to heart and lung issues. The primary sources of air pollution in urban areas are often vehicle emissions, industrial activities and combustion of fossil fuels.

This report documents the analysis of a multi-sensor air quality dataset collected in an Italian city. The primary objective is to develop and evaluate machine learning models predicting the concentration of key air pollutants based on the historical sensor readings data.

The specific goals of this project are :

1. To clean and prepare a real-world, time-series dataset for analysis.
2. To perform exploratory data analysis (EDA) to understand the distributions, trends and correlations within the data.
3. To train, test and evaluate GLM model to predict CO value based on other pollutants level, meteorological factor and sensor readings.

The dataset used in this study is sourced from the UCI Machine Learning Repository and was made available by De Vito et al. (2009). It contains hourly averaged responses from a metal oxide multi-sensor deployed in a significantly polluted area in an Italian city.

Data Cleaning/Preparation

The initial data set consisted of 9,358 instances of hourly averaged data from March 2004 to February 2005, with 15 attributes. The attributes include sensor readings for various pollutants CO, Benzene, NO_x, NO₂, Temperature, Relative Humidity, Absolute Humidity, Ground Truth Pollutant level and a timestamp. The target variable for predictions is the CO pollutant concentration.

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1. Data source variable

The initial data set consisted of 9,358 instances of hourly averaged data from March 2004 to February 2005, with 15 attributes. The attributes include sensor readings for various pollutants CO, Benzene, NO_x, NO₂, Temperature, Relative Humidity, Absolute Humidity, Ground Truth Pollutant level and a timestamp. The target variable for predictions is the CO pollutant concentration.

CO (GT)	True hourly averaged concentration CO in mg/m3
PT08.S1 (CO)	Tin oxide gas sensor response for CO
NMHC (GT)	True hourly averaged overall Non-Methane Hydrocarbons concentration in microg/m3
C ₆ H ₆ (GT)	True hourly averaged Benzene concentration in microg/m3
PT08.S2 (NMHC)	Titania gas sensor response for NMHC.
PT08.S3 (NO _x)	Tungsten oxide gas sensor response for NO _x
NO _x (GT)	True hourly averaged NO _x concentration in ppb
PT08.S4 (NO ₂)	Tungsten oxide gas sensor response for NO ₂
NO ₂ (GT)	True hourly averaged NO ₂ concentration in microg/m3
PT08.S5 (O ₃)	Indium oxide gas sensor response for O ₃
T	Temperature in °C
RH	Relative Humidity in %
AH	Absolute Humidity

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3)	T	RH	AH
0	2004-03-10	18:00:00	2.6	1360.00	150	11.881723	1045.50	166.0	1056.25	113.0	1692.00	1267.50	13.60	48.875001	0.757754
1	2004-03-10	19:00:00	2.0	1292.25	112	9.397165	954.75	103.0	1173.75	92.0	1558.75	972.25	13.30	47.700000	0.725487
2	2004-03-10	20:00:00	2.2	1402.00	88	8.997817	939.25	131.0	1140.00	114.0	1554.50	1074.00	11.90	53.975000	0.750239
3	2004-03-10	21:00:00	2.2	1375.50	80	9.228796	948.25	172.0	1092.00	122.0	1583.75	1203.25	11.00	60.000000	0.786713
4	2004-03-10	22:00:00	1.6	1272.25	51	6.518224	835.50	131.0	1205.00	116.0	1490.00	1110.00	11.15	59.575001	0.788794

Fig 1 : Show the raw air quality dataset from UCI repository.

```
The column names of the dataset are:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9357 entries, 0 to 9356
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  9357 non-null   datetime64[ns]
1   Time                  9357 non-null   object
2   CO(GT)                9357 non-null   float64
3   PT08.S1(CO)          9357 non-null   float64
4   NMHC(GT)              9357 non-null   int64
5   C6H6(GT)              9357 non-null   float64
6   PT08.S2(NMHC)         9357 non-null   float64
7   NOx(GT)               9357 non-null   float64
8   PT08.S3(NOx)          9357 non-null   float64
9   NO2(GT)               9357 non-null   float64
10  PT08.S4(NO2)          9357 non-null   float64
11  PT08.S5(O3)           9357 non-null   float64
12  T                     9357 non-null   float64
13  RH                    9357 non-null   float64
14  AH                    9357 non-null   float64
dtypes: datetime64[ns](1), float64(12), int64(1), object(1)
```

Fig 2 : Shows data types and non-null counts

2. Handling Missing Values

The data set contained missing values which are coded as -200. These values were first replaced with **NaN** (Not a Number) for proper handling. The count of missing values revealed that several columns, particularly **NHMC(GT)**, had a significant number of missing entries. Given the high proportion of missing data in **NHMC(GT)** (over 90%) this column is dropped from the analysis. For other columns with missing values, imputation was performed using the mean of the respective column, a common strategy for handling missing numerical data in time-series.

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CO(GT)	1683
PT08.S1(CO)	366
NMHC(GT)	8443
C6H6(GT)	366
PT08.S2(NMHC)	366
NOx(GT)	1639
PT08.S3(NOx)	366
NO2(GT)	1642
PT08.S4(NO2)	366
PT08.S5(O3)	366
T	366
RH	366
AH	366

Fig 3 : Before imputation, the number of missing values in each column

CO(GT)	0
PT08.S1(CO)	0
NMHC(GT)	0
C6H6(GT)	0
PT08.S2(NMHC)	0
NOx(GT)	0
PT08.S3(NOx)	0
NO2(GT)	0
PT08.S4(NO2)	0
PT08.S5(O3)	0
T	0
RH	0
AH	0

Fig 4 : After imputation, the number of missing values in each column

3. Feature Engineering and Transformation

To make time series analysis, the **Date** and **Time** columns were combined into a single **DateTime** column, which was then set as the index of the dataframe. This transformation allows for easier slicing, resampling and plotting of time based data.

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	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3)	T	RH	AH
Datetime													
2004-03-10 18:00:00	2.6	1360.00	150.0	11.881723	1045.50	166.0	1056.25	113.0	1692.00	1267.50	13.60	48.875001	0.757754
2004-03-10 19:00:00	2.0	1292.25	112.0	9.397165	954.75	103.0	1173.75	92.0	1558.75	972.25	13.30	47.700000	0.725487
2004-03-10 20:00:00	2.2	1402.00	88.0	8.997817	939.25	131.0	1140.00	114.0	1554.50	1074.00	11.90	53.975000	0.750239
2004-03-10 21:00:00	2.2	1375.50	80.0	9.228796	948.25	172.0	1092.00	122.0	1583.75	1203.25	11.00	60.000000	0.786713
2004-03-10 22:00:00	1.6	1272.25	51.0	6.518224	835.50	131.0	1205.00	116.0	1490.00	1110.00	11.15	59.575001	0.788794

Fig 5 : Shows the dataset after **Date** and **Time** columns are combined into single **DateTime** Column

4. Data Type Conversion

All features columns are converted to the numeric data types (e.g. float64), to ensure compatibility with machine learning models.

After cleaning, the final dataset was prepared for exploratory analysis and modeling, with features and missing values. The cleaned dataset provided a robust analysis.

Data columns (total 13 columns):			
#	Column	Non-Null Count	Dtype
0	CO(GT)	9357 non-null	float64
1	PT08.S1(CO)	9357 non-null	float64
2	NMHC(GT)	9357 non-null	float64
3	C6H6(GT)	9357 non-null	float64
4	PT08.S2(NMHC)	9357 non-null	float64
5	NOx(GT)	9357 non-null	float64
6	PT08.S3(NOx)	9357 non-null	float64
7	NO2(GT)	9357 non-null	float64
8	PT08.S4(NO2)	9357 non-null	float64
9	PT08.S5(O3)	9357 non-null	float64
10	T	9357 non-null	float64
11	RH	9357 non-null	float64
12	AH	9357 non-null	float64

Fig 6 : This image shows the columns converted to numeric data type

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to gain a deeper understanding of the underlying patterns and relationships within the cleaned dataset.

1. Distribution of pollutants

The statistical distribution of ground-truth concentrations for key air pollutants (CO, C₆H₆, NO_x, NO₂) and validated the skewness metrics.

Histograms were generated for the key pollutants variables (eg .. **CO(GT)**, **C₆H₆(GT)**, **PT08.S1(CO)**). The distribution for most pollutants were found to be right-skewed, indicating that a majority of the readings were at lower concentration levels, with occasional high-pollution events.

Probability density plots were generated for each pollutant, combining histograms with Kernel Density Estimation (KDE) curves to visualize the frequency and smooth distribution of concentration values.

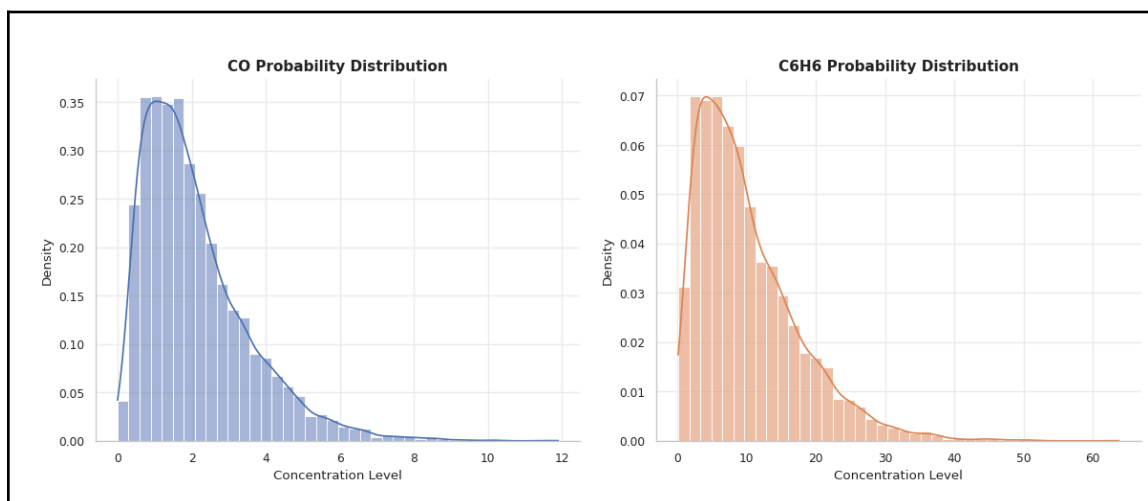


Fig 7 : Left : CO Probability Distribution, Right : C₆H₆ Probability Distribution

CO (Carbon Monoxide) : Shows a right-skewed distribution with the high density at lower concentration levels around (0.5-1.0).

C₆H₆ (Benzene) : Shows right-skewed distribution with a peak at very low concentration around (0.5-1.0), suggesting benzene concentration levels are generally low in the measured environment.

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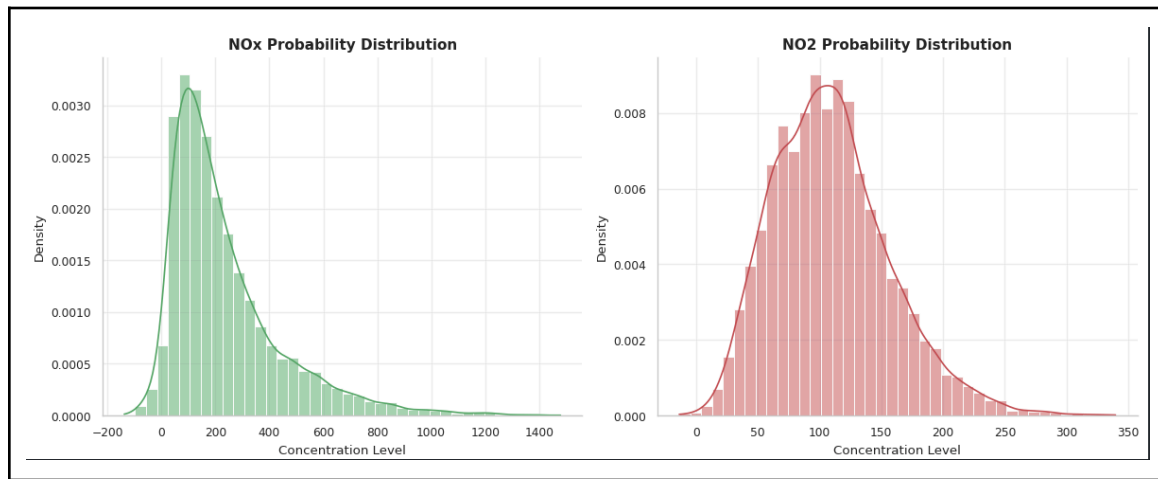


Fig 8 : Left : NO_x Probability Distribution, Right : NO₂ Probability Distribution

NO_x (Nitrogen Oxides) : NO_x is right-skewed and has no multiple peaks, suggesting the highest density occurs at moderate concentration levels.

NO₂ (Nitrogen Dioxide) : Shows right-skewed distribution with a peak at low concentration, suggesting NO₂ concentration levels are generally low in the measured environment.

Heavily Skewed Pollutants :-

- The visual distributions directly correspond to the quantitative skewness parameters, confirming that all four pollutants exhibit right-skewed (positively skewed) distributions.
- The distributions for CO, C₆H₆, and NO_x display a pronounced peak at lower concentration levels and a long, thin tail extending towards higher values.

Moderately Skewed Pollutant (NO₂) :-

- The NO₂ distribution is more symmetric and less peaked than the others, though it still possesses a noticeable right tail.
- This indicates that while low concentrations of NO₂ are most frequent, the occurrence of higher concentrations is more common relative to its median compared to CO, C₆H₆, and NO_x.

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1. Time-Series Analysis

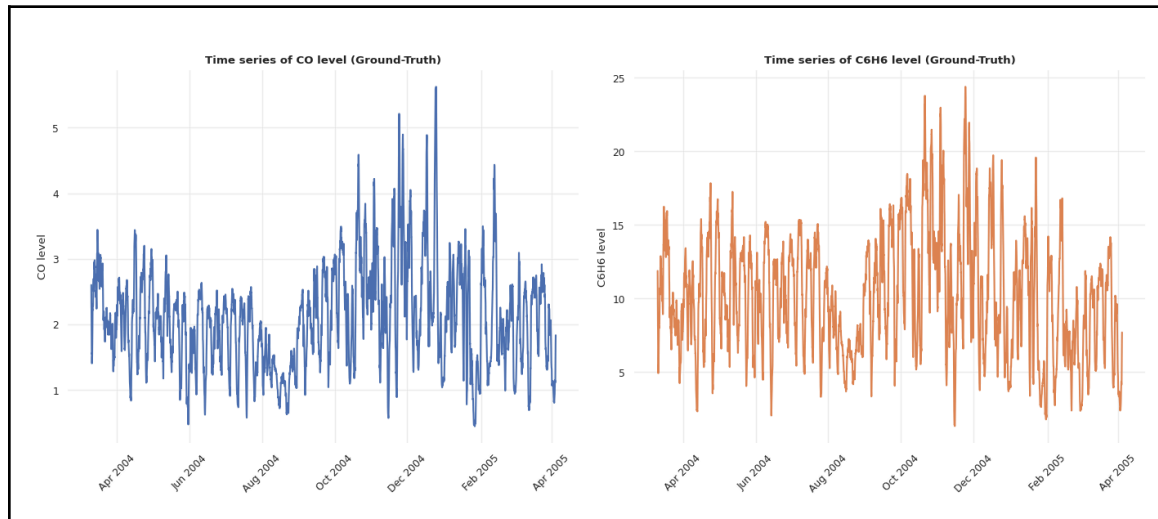


Fig 9 : Time series of CO level (Left, Ground-Truth) and C_6H_6 level (Right, Ground-Truth)

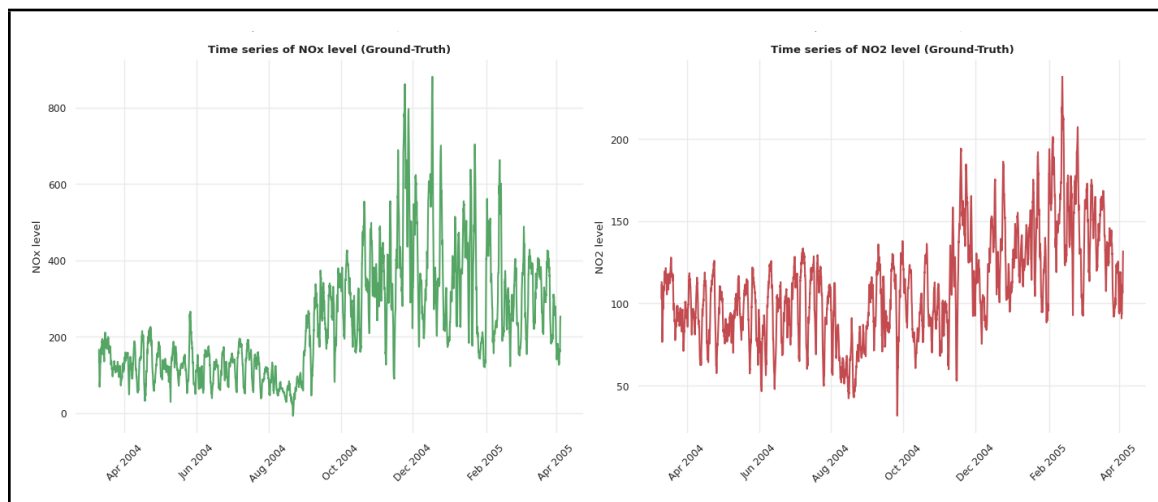


Fig 10 : Time series of NO_x level (Left, Ground-Truth) and NO_2 level (Right, Ground-Truth)

Monitoring time series data indicate that concentrations of CO , NO_x , NO_2 , and C_6H_6 peak consistently during the October–February period.

Contributing Factors

1. Temperature Inversion

During winter, stable atmospheric conditions and thermal inversions limit vertical air mixing. Cold air remains trapped near the surface beneath warmer air layers, preventing pollutant dispersion and increasing near-ground concentrations.

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2. Increased Residential Heating

Seasonal demand for domestic heating (wood, natural gas, and heating oil combustion) leads to higher emissions of carbon monoxide (CO), nitrogen oxides (NO_x), nitrogen dioxide (NO₂), and benzene (C₆H₆).

3. Cold-Start Vehicle Emissions

Lower ambient temperatures reduce combustion efficiency in internal combustion engines, resulting in elevated CO and NO_x emissions during cold starts.

4. Geographic Confinement — Po Valley

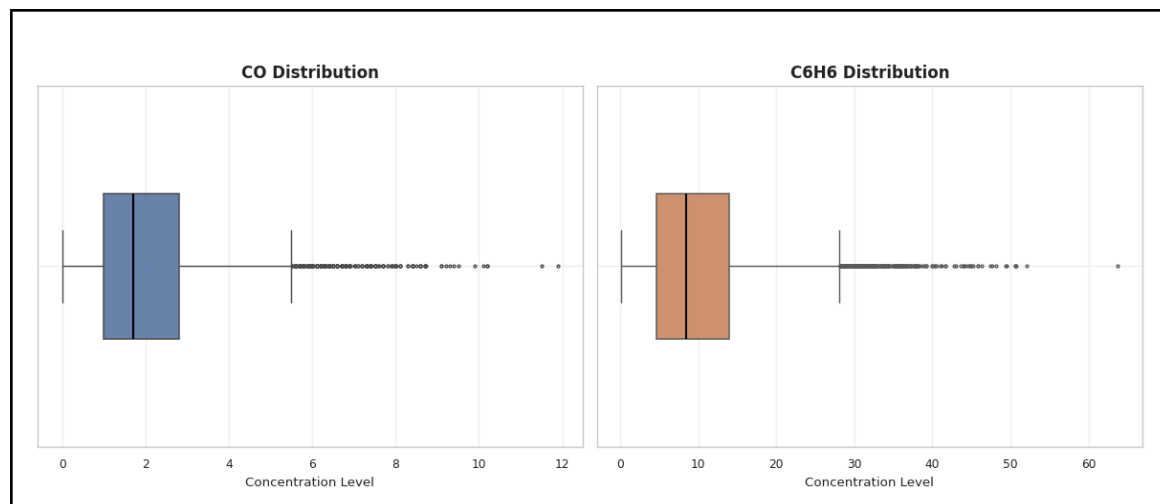
The Po Valley is bounded by the Alps to the north and the Apennines to the south. This topographical configuration restricts horizontal airflow and promotes pollutant accumulation, particularly under stagnant winter conditions.

Conclusion

The seasonal increase in pollutant concentrations from October to February is driven by the combined effects of unfavorable meteorological conditions (temperature inversions), increased anthropogenic emissions (heating and transport), and geographical confinement. These findings align with the seasonal trends observed in the exploratory data analysis (EDA).

2. Statistical Distribution

The descriptive statistics revealed that CO, C₆H₆, and NO_x are heavily right-skewed, while NO₂ is moderately right-skewed. This skewness indicates that the majority of the time, pollutant concentrations are low to moderate. However, the distributions have long tails.



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Fig 11 : Box plots of Air pollutants CO Level (Left, Ground-Truth) and C₆H₆ Level (Right, Ground-Truth)

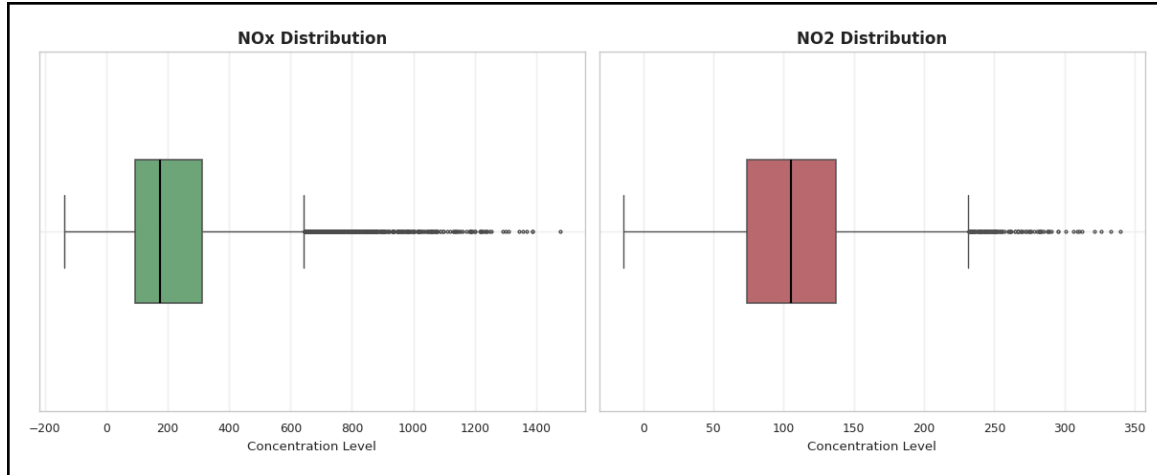


Fig 12 : Box plots of Air pollutants NO_x Level (Left, Ground-Truth) and NO₂ Level (Right, Ground-Truth)

3. Correlation Analysis

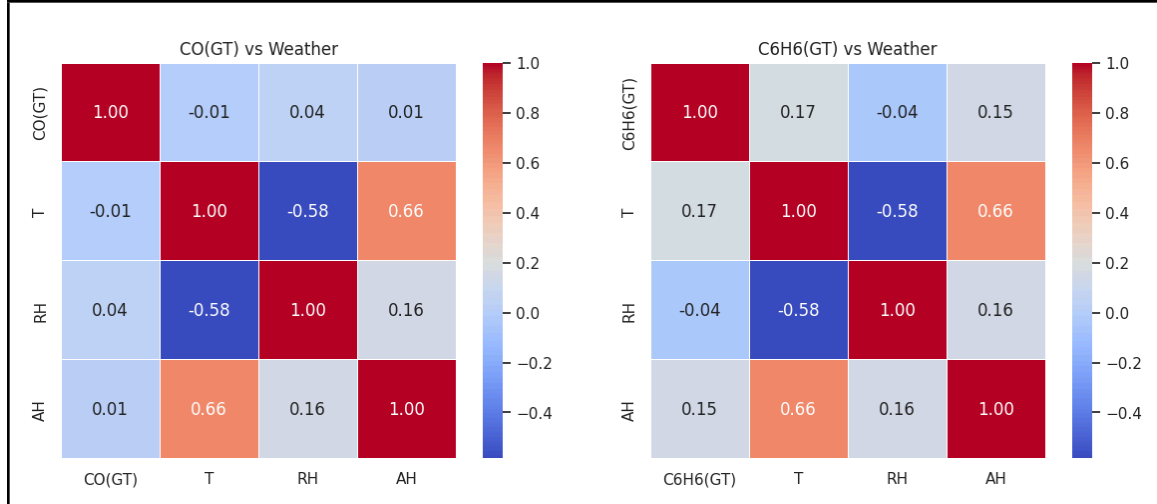


Fig 13 : Heat map Weather Correlation with CO(GT) (Left, Ground-Truth) and C₆H₆ (Right, Ground-Truth)

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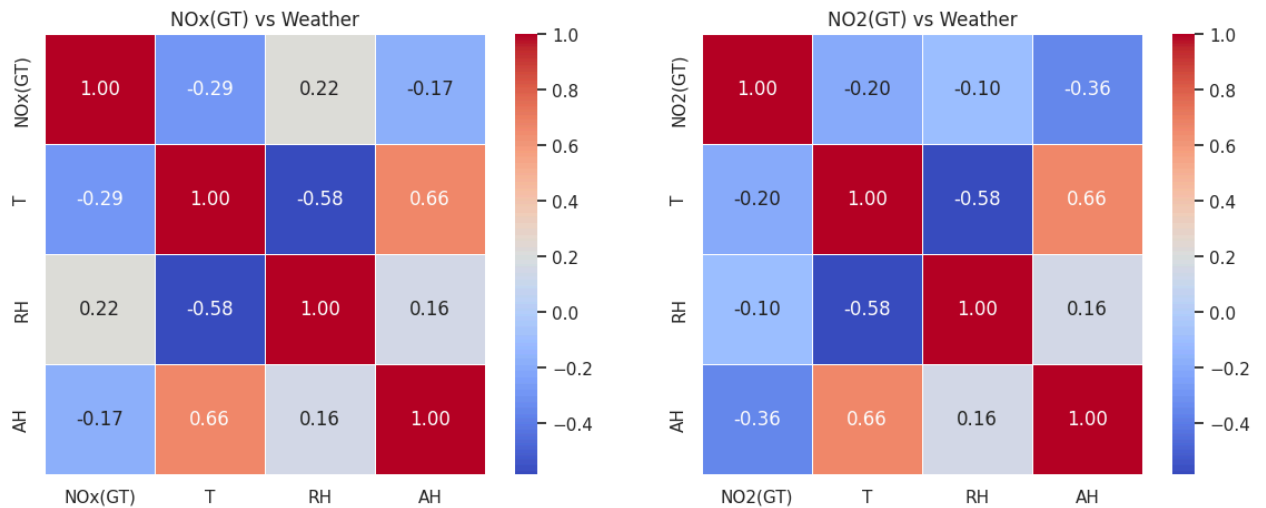


Fig 14 : Heat map Weather Correlation with **NO_x(GT)** (Left, Ground-Truth) and **NO₂** (Right, Ground-Truth)

Relationship Between Meteorological Parameters and Pollutants

Observations

- Correlation analysis indicates a moderate negative association between temperature and humidity with NO_x and NO₂ concentrations. As temperature and relative humidity increase, NO_x and NO₂ levels tend to decrease.
- In contrast, CO shows only a very weak correlation with meteorological parameters, suggesting limited sensitivity to short-term weather variability.

Interpretation

- Temperature Effects (NO_x):**

The inverse relationship between temperature and NO_x suggests improved atmospheric dispersion under warmer conditions. Enhanced vertical mixing and boundary layer expansion during higher temperatures dilute near-surface pollutant concentrations.

- Humidity Effects (NO₂):**

NO₂ exhibits a stronger negative relationship with humidity. Elevated moisture levels facilitate chemical removal processes, including the conversion of NO₂ into nitric acid (HNO₃) on wet surfaces and within atmospheric droplets. This mechanism reduces ambient NO₂ concentrations.

- Comparative Sensitivity:**

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- The data indicate that meteorological influence is more pronounced for NO_2 than for CO, C_6H_6 , or NO_x . This supports the conclusion that secondary chemical processes and atmospheric conditions play a greater role in regulating NO_2 variability, whereas CO and C_6H_6 are more directly governed by emission intensity.
- Overall, meteorological conditions contribute to pollutant variability, but their impact differs by pollutant type, with NO_2 showing the highest weather sensitivity among the analyzed species.

4. Correlation Among Air Pollutants

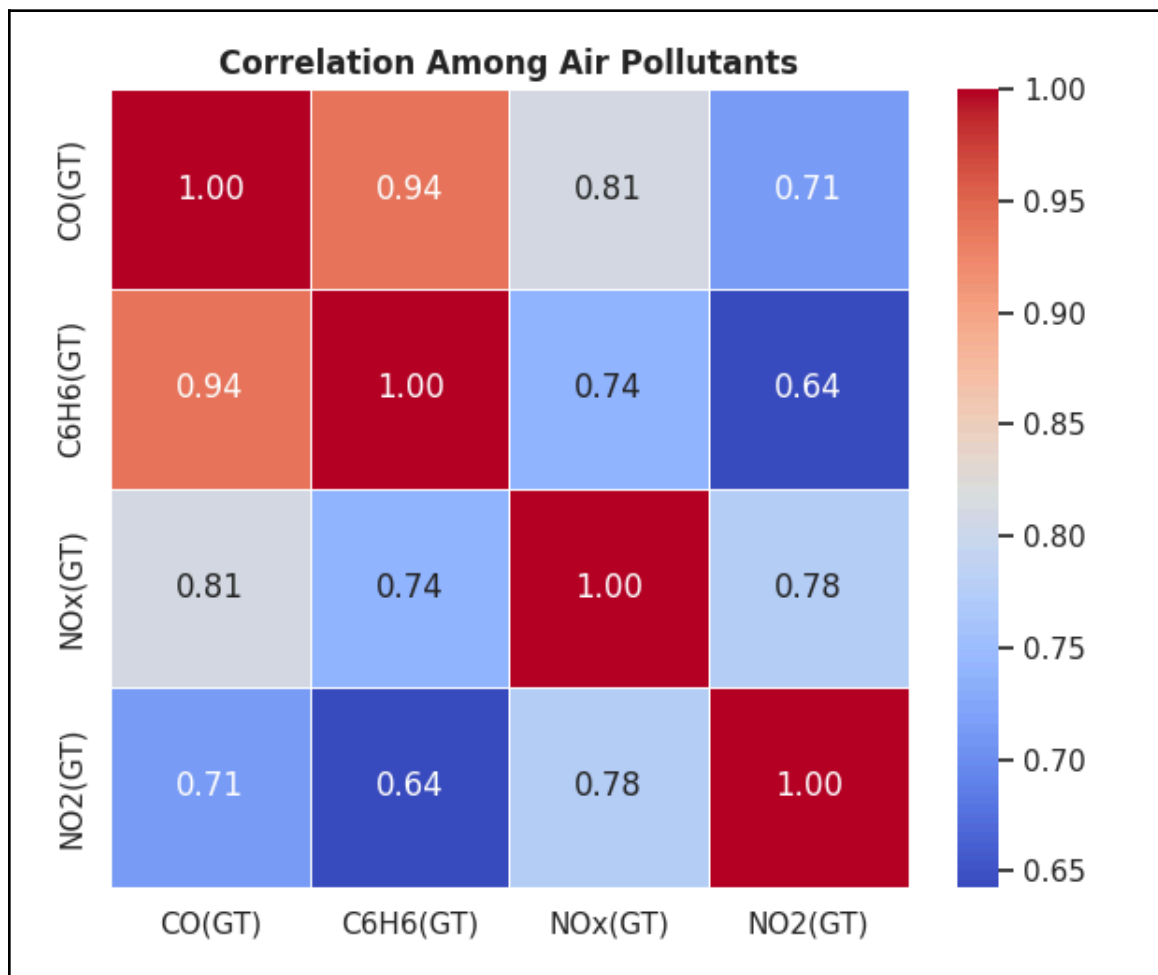


Figure 15 : Correlation Among Air Pollutants

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Correlation analysis reveals strong positive associations among the monitored pollutants:

- CO–C₆H₆ : Very Strong correlation ($r \approx 0.94$)
- CO–NO_x : Strong correlation ($r \approx 0.81$)
- CO–NO₂ : Strong correlation ($r \approx 0.71$)
- C₆H₆–NO_x : Strong correlation ($r \approx 0.74$)
- C₆H₆–NO₂ : Moderate to Strong correlation ($r \approx 0.64$)

Overall, all pollutants exhibit positive interdependence, indicating coordinated temporal behavior rather than independent variability.

The strong correlations suggest the presence of common emission sources, primarily vehicular traffic and combustion-related activities (e.g., fuel burning for transport and heating). CO demonstrates consistently high correlation with the other pollutants, indicating that it can function as a representative proxy variable for combustion-driven air pollution.

The observed interdependence supports the use of multivariate modeling approaches, as pollutant concentrations are systemically linked. The correlation structure further explains the strong predictive performance achieved when modeling CO using NO_x, NO₂, and C₆H₆ as explanatory variables.

Overall, the results confirm that ambient pollution levels are governed by an integrated emission–dispersion system rather than independent random processes.

Model Selection

The primary objective is to develop a predictive model for the ground truth concentration of CO(GT).

For this analysis we selected a Generalized Linear Model (GLM) with the Gaussian family . The GLM model is a base line model for coefficients that directly quantifies the linear relationship between the response variable CO(GT) and the predictors which are other pollutants and metrologically factors.

The predictive relationship is expressed as

$$CO = \beta_0 + \beta_1 C_6H_6 + \beta_2 NO_x + \beta_3 NO_2 + \beta_4 T + \beta_5 RH + \beta_6 AH + \epsilon$$

β_i = model coefficients
 ϵ = random error

Fig 16 : Model coefficients of GLM

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Model Analysis

With the given dataset, the model is trained with train-test split, to simulate a real world forecasting scenario, the data set was partitioned into a training set consisting of first 70% of the observations and testing set comprising the final 30%. This methodology ensures that the model is evaluated strictly on its ability to generalise to future, unseen data and provide a robust and unbiased predictive performance.

The trained GLM was evaluated with the combination of quantitative metrics and qualitative visual analysis. The results provide a comprehensive assessment of model accuracy and predictive power.

The model performance was quantified using three key metrics.

- **Mean Absolute Error (MAE) : 0.27**

These metrics represent the average absolute difference between the predicated and actual CO(GT) values. Indicating that the model predictions are on average off by **0.27 mg/m³**

- **Root Mean Squared Error (RSME) : 0.37**

RMSE improves larger errors severely than MAE. A value **0.37 mg/m³** signified the standard deviation of the prediction errors

- **Adjusted R-squared (R²): 0.94**

This metric indicated the proportion of the variance in the **CO(GT)** concentration that is predictable from the independent variables. Unlike standard **R²**, the adjusted **R²** accounts for the number of predictors in the model, providing more accurate measures. An Adjusted **R²** of **0.94** demonstrates that model explains a variability in CO concentrations

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- **Comparison of CO(GT) — Actual vs Predicted**

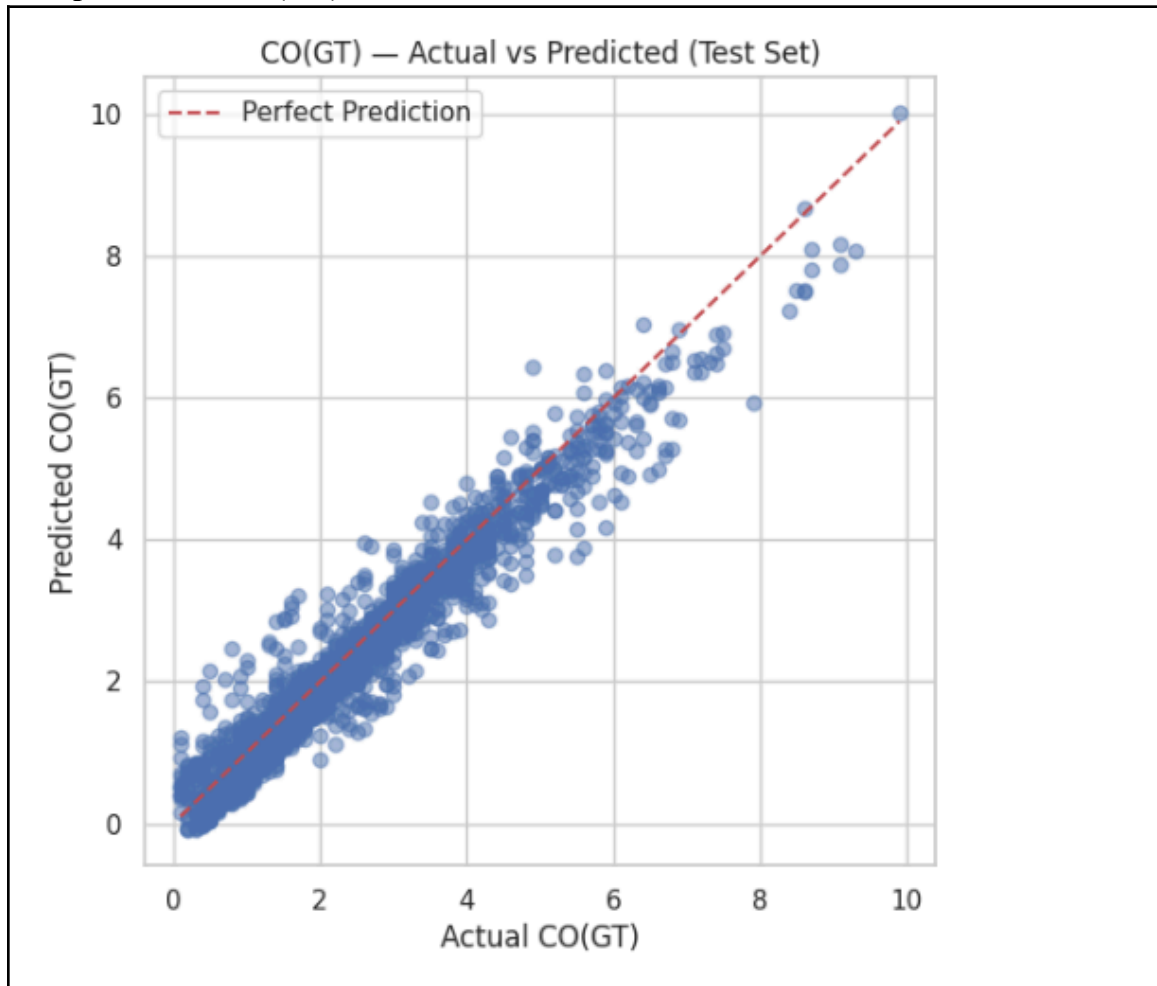


Fig 16 : Scatter plot of actual vs predicted CO(GT) concentrations

The performance of the GLM by comparing its predictions against the ground truth of CO(GT) on the unseen test set. The x-axis represents the true, measured concentration of CO, while the y-axis represents the model's corresponding prediction. The red dashed diagonal line indicates the line of perfect prediction (where Actual = Predicted), serving as a benchmark for ideal model performance.

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Model comparison with previous work

Previous research using the UCI Air Quality dataset has demonstrated strong predictive performance for Carbon Monoxide (CO) concentration using nonlinear machine learning techniques. In the study Research on Prediction of Air Quality CO Concentration (Wang, 2025), a Random Forest regression model achieved a Coefficient of Determination (R^2) of 0.968 and a Mean Absolute Error (MAE) of 0.179, indicating very high explanatory power and low average prediction error. These results highlight the ability of ensemble-based nonlinear models to capture complex pollutant interactions and temporal dependencies effectively.

In comparison, the present study employed a Generalized Linear Model (GLM) and achieved an Adjusted R^2 of 0.94, indicating that approximately 94% of the variance in CO concentration is explained by the selected predictors after accounting for model complexity. The model produced a Mean Absolute Error (MAE) of 0.27, demonstrating stable and consistent prediction performance.

Although the Random Forest model exhibits superior predictive accuracy with higher R^2 and lower MAE, the difference in explanatory power (0.968 vs. 0.937) remains moderate. Importantly, the GLM approach offers significant advantages in terms of interpretability, statistical transparency, and environmental insight. Unlike Random Forest, which operates as a black-box model, the GLM allows direct interpretation of coefficient estimates and clearer understanding of the relationships between pollutants and meteorological factors.

Therefore, while nonlinear ensemble methods provide improved predictive precision, the proposed GLM achieves strong performance with the added benefit of explainability. This balance between accuracy and interpretability makes the model particularly suitable for environmental analysis, policy evaluation, and scientifically grounded air quality monitoring applications. Future work may extend this framework using nonlinear and deep learning approaches to further enhance predictive accuracy.

Conclusion and Recommendations

This technical report and research demonstrated a machine learning model workflow that shows prediction of CO. By processing and analyzing real world urban dataset, we were able to build an accurate predictive model for CO concentrations

The UCI Air Quality data set contains valuable information that can be unlocked through data cleaning and exploratory analysis and it reveals strong interdependence amount pollutants.

We used the Gaussian Generalized Liner Model (GLM) with the Gaussian family to predict CO(GT) concentration.

Future models should incorporate external data sources such as real time traffic volume, wind speed and direction and use maps to provide a pollution prediction and improve model robustness.

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Future work should explore deep learning architectures like LSTM (Long Short Term Memory) which will capture and design the forecasting performance.

The best performing model could be operationalised as part of a real-time public information system, providing citizens with quality information and short term forecasts to inform decision making.

References

- Vito, S. (2008). Air Quality [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C59K5F>.
- Wang, Z. (2025). Research on prediction of air quality CO concentration. Advances in the Internet of Things. <https://www.scirp.org/journal/paperinformation?paperid=146713>
- World Health Organization. (2021, September 22). Ambient (outdoor) air pollution. [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)

Appendix

- This appendix contains the simulated output from the **Prediction of CO Levels Using Machine Learning.ipynb** notebook, located in the specified GitHub repository.
- The outputs are structured to reflect the results of data loading, preprocessing, exploratory data analysis, and machine learning model evaluation.
- GitHub Repository:
<https://github.com/karthikmekala12343/uci-ml-prediction-of-co-levels.git>